Later it was identified that there were many substances which retarded the speed of a reaction.

Hence a catalyst is defined as a substance which alters the rate of chemical reaction without itself undergoing chemical change. The phenomenon which involves the action of a catalyst is called catalysis.

Positive and negative catalysis:

In positive catalysis, the rate of a reaction is increased by the presence of catalyst but in negative catalysis, the rate of reaction is decreased by the presence of a catalyst.

The two main types of catalysis (i) Homogeneous catalysis and (ii) Heterogeneous catalysis

Homogeneous catalysis

In a homogeneous catalysed reaction, the reactants, products and catalyst are present in the same phase.

Illustration (1):

$$2SO_{2(g)} + O_{2(g)} + [NO]_{(g)} \rightarrow 2SO_{3(g)} + [NO]_{(g)}$$

In this reaction the catalyst NO, reactants, SO_2 and O_2, and product, SO_3 are present in the gaseous form.

Illustration (2):

In the decomposition of acetaldehyde by I_2 catalyst, the reactants and products are all present in the vapour phase.

$$CH_3CHO_{(g)} + [I_2]_{(g)} \rightarrow CH_{4(g)} + CO_{(g)} + [I_2]_{(g)}$$

Let us consider some examples in which the reactants, products and catalyst are present in aqueous solution.

(1) Hydrolysis of cane sugar with a mineral acid as catalyst

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11(l)}} + H_2O_{(l)} \xrightarrow{H_2SO_{4(l)}} \underset{\text{Glucose}}{C_6H_{12}O_{6(l)}} + \underset{\text{Fructose}}{C_6H_{12}O_{6(l)}}$$

(2) Ester hydrolysis with acid or alkali as catalyst

$$\underset{\text{Ethyl acetate}}{CH_3COOC_2H_{5(l)}} + H_2O_{(l)} \xrightarrow{H_2SO_{4(l)}} \underset{\text{acetic acid}}{CH_3COOH_{(l)}} + \underset{\text{ethyl alcohol}}{C_2H_5OH_{(l)}}$$

Heterogeneous catalysis

In a reaction, the catalyst is present in a different phase i.e. it is not present in the same phase as that of reactants or products. This is generally referred as contact catalysis and the catalyst present is in the form of finely divided metal or as gauze

Illustration

(i) In the manufacture of sulphuric acid by contact process SO_3 is prepared by the action of SO_2 and O_2 in the presence of Pt or V_2O_5 as a catalyst.

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